Environmental Health Risk Model: Data Exploration and Optimization

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Data Exploration and Analysis

Distribution of Key Variables

The three variables identified to be analyzed are temperature, dew, and PM2.5 particulate matter. Statistical analysis was utilized to characterize their distributions and assess their potential effects on the target variable. See Table 1 for key measurements found during the analysis.

Feature	Corr	Mean	Median	StdDev	Min	Max	Skewness	Kurtosis	IQR
Temp	0.41	76.12	75.22	8.72	55.55	99.85	0.35	-0.52	13.3
PM2.5	0.29	32.74	32.7	6.31	11.74	52.38	-0.07	-0.11	8.93
Dew	0.58	57.27	58.6	9.16	26.26	76.65	-0.39	0.05	11.51

Table 1. Temperature, PM2.5, and Dew and related statistical measures.

Temperature (Figure 1) shows a slight right skew of 0.35 and light tails (kurtosis -0.52), indicating that it is balanced with few extremes. The mean of 76.12 is greater than the median of 75.22, which matches the slight right tail.

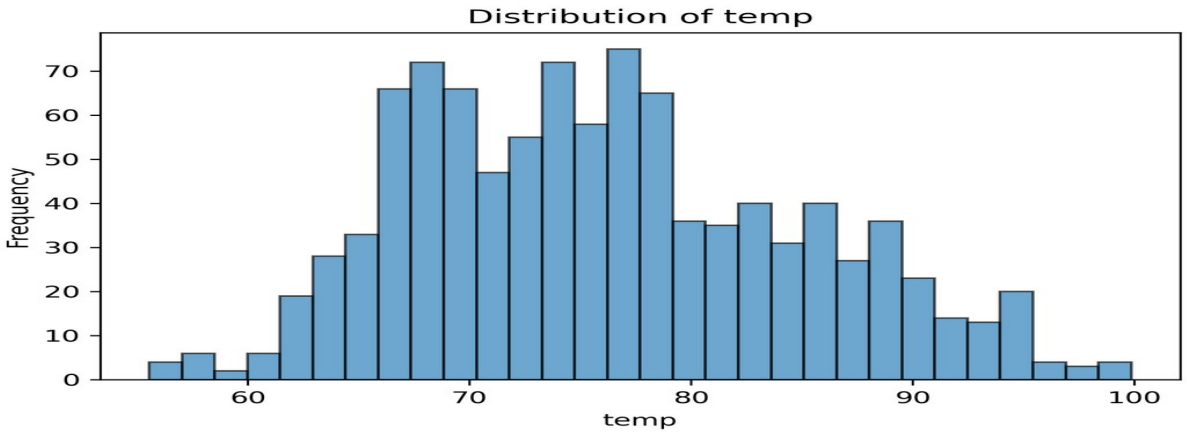


Figure 1. Histogram of daily temperature values (°F).

PM 2.5 (Figure 2) shows a symmetric skew of -0.07 with near normal tails (-0.11). Mean and median are approximately the same, with 32.74 and 32.70, respectively, indicating a clean, well-behaved variable for analysis.

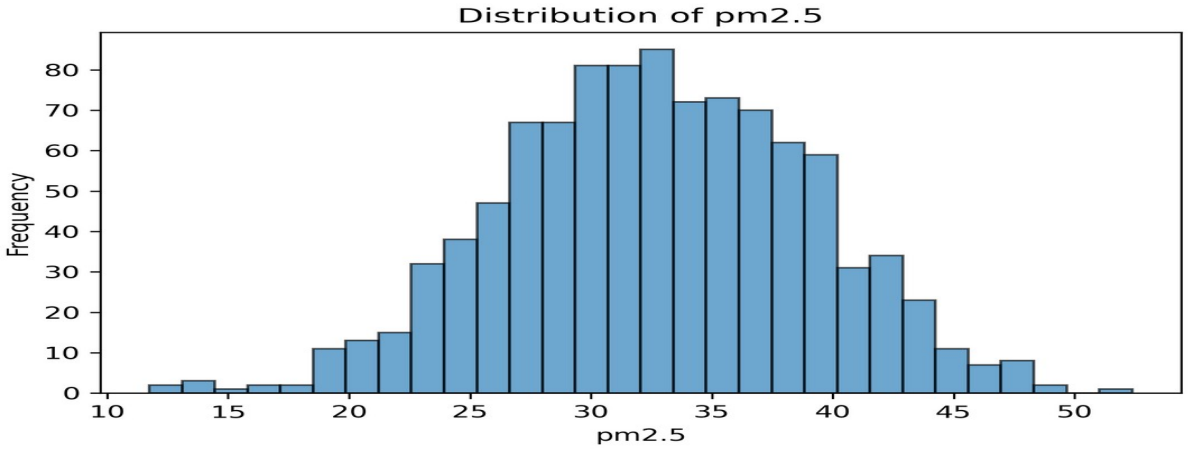


Figure 2. Histogram of daily PM 2.5 concentrations (μg m^{-3}).

Dew (Figure 3) shows a mild left skew (-0.39) with near normal tails (0.05). The median is larger than the mean at 58.6 and 57.27, respectively, consistent with the small left tail shown.

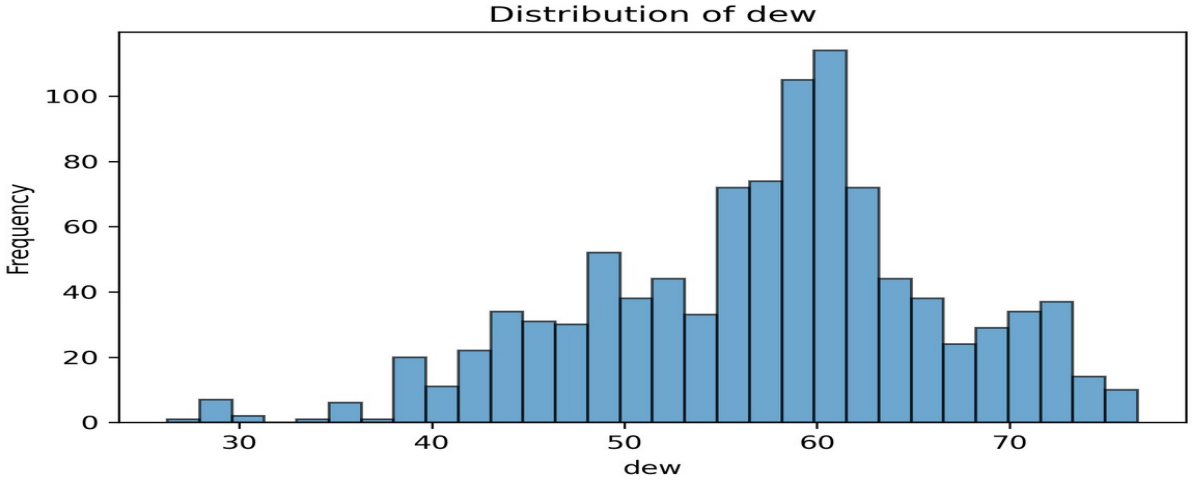


Figure 3. Histogram of daily Dew.

Underlying Patterns

1. **Moist Heat Dominates Risk**

   Dew and heat-related measures have the strongest linear associations with the target. The evaluation of LOWESS curves for each heat and moisture feature shows a strong correlation for significant increases in health risk scores. See Figures 4 through 6. The flattening of the overall risk scores can be attributed to decreased activity outdoors due to human behavior. When extreme moist heat is present, health risks flatten out or stay the same because humans tend to stay indoors. It is higher in the ranges of highest risk scores because humans tend to gravitate towards the outdoors when there is more “favorable” weather.

2. **Particulate Matter 2.5 adds signal but is secondary to meteorology**

   PM2.5 correlates moderately with the target (r ≈0.29) but shows only minor attributions to the model with heat and moisture features. See Figure 7. Dew and Heat Index remain the highest overall contributors towards health risk.

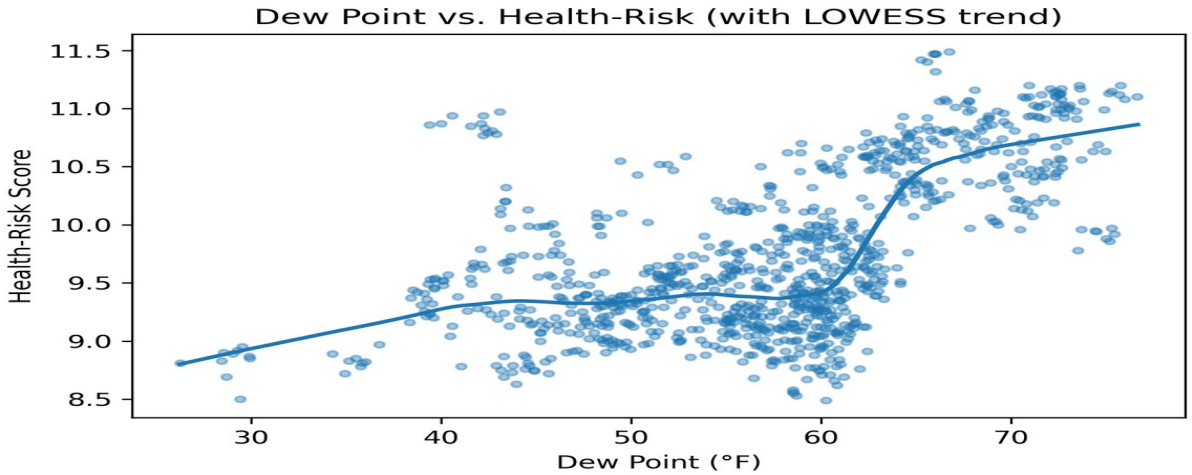


Figure 4. Scatterplot of Dew Point versus Health-Risk Score with LOWESS smoother.

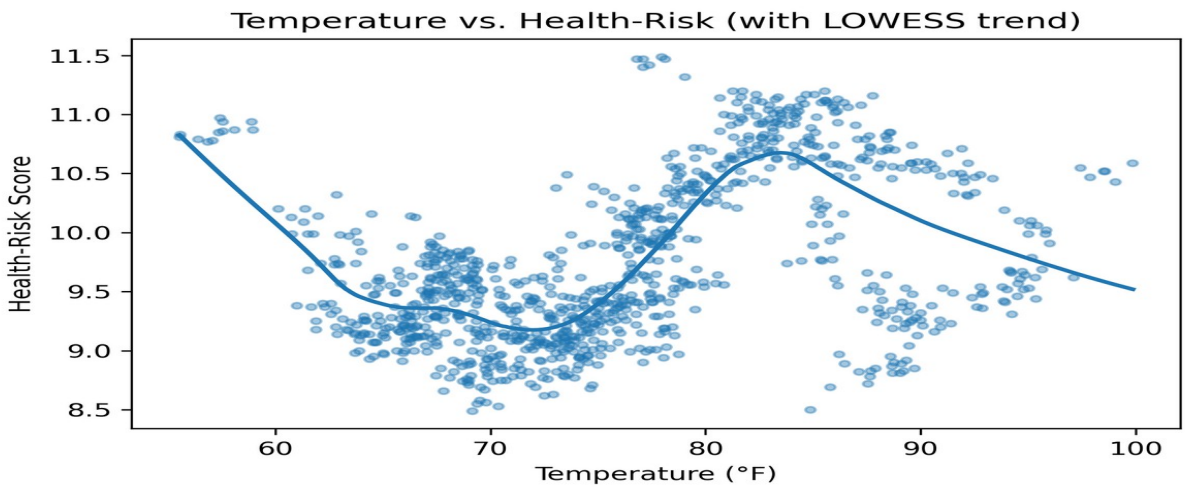


Figure 5. Scatterplot of Temperature versus Health-Risk Score with LOWESS smoother.

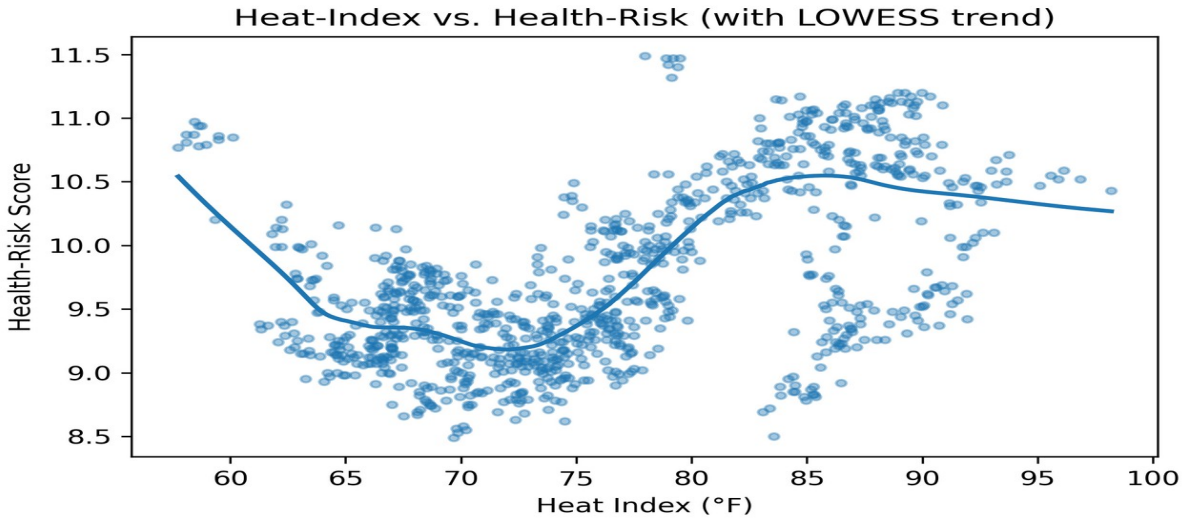


Figure 6. Scatterplot of Heat Index versus Health-Risk Score with LOWESS smoother.

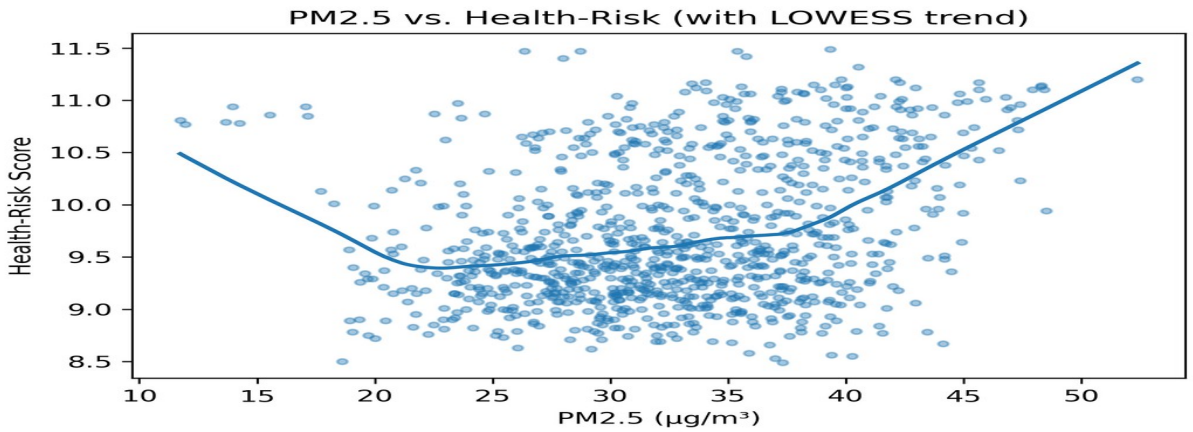


Figure 7. Scatterplot of PM2.5 versus Health-Risk Score with LOWESS smoother.

Hypothesis

Due to an evident increase in health risk associated with both temperature and moisture features, it is hypothesized that these metrics would be the most significant overall indicator of health risk. When dew ≥ ~65 and temperature ≥ ~85 (or heat index ≥ ~80–85), health risk rises disproportionately, independent of moderate PM2.5 fluctuations. When both are at high levels, these metrics indicate a higher risk to people. See Figure 8.

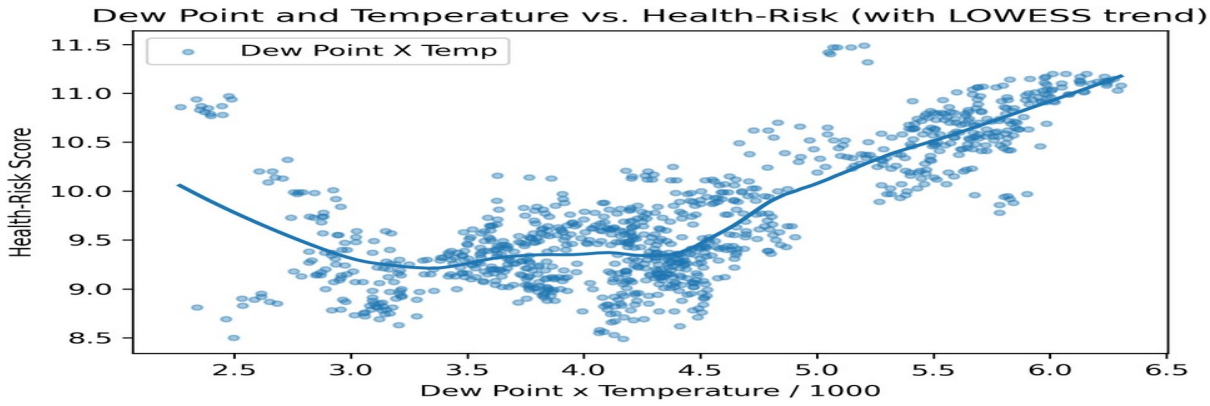


Figure 8. Scatterplot of Dew X Temp versus Health-Risk Score with LOWESS smoother.

Interpretation of Model Outputs

Performance Metrics

The optimized random-forest regressor achieved RMSE of 0.108 and R-squared of 0.974 on the hold-out set. This outperformed a regularized linear baseline RMSE of 0.115 and R-squared of 0.971. Cross-validation shows RMSE = 0.1106 ± 0.008 and R-Squared = 0.9733 ± 0.0042. This is a strong indicator of great generalization. With these metrics, the model is suitable for operational forecasting and threshold-based alerts, provided there is routine monitoring for distribution shifts (i.e., extreme smoke events).

Model Behavior Insights

1. **Dew is the dominant driver.**

   SHAP shows high-dew observations (red points) pushing predictions upward with a wide positive spread; Random-Forest importances concur. This supports using dew-based early-warning rules. The engineered feature of Dew X Temperature could be an even potentially stronger candidate for modeling. See Figures 9 and 10.
2. **Heat-index curvature and threshold.**

   The LOWESS shape confirms a regime switch: relatively benign below ~80, rapidly increasing risk from ~80–90, then a plateau. This nonlinearity is precisely what tree-based models capture well and should be reflected in decision thresholds.
3. **PM2.5 is conditionally modest.**

   Narrow SHAP spread and absence from top importances suggest its marginal contribution is minor within this PM2.5 range. It is still advised to keep the feature in the

model as its influence is expected to grow during high-variance episodes (e.g., wildfire smoke).

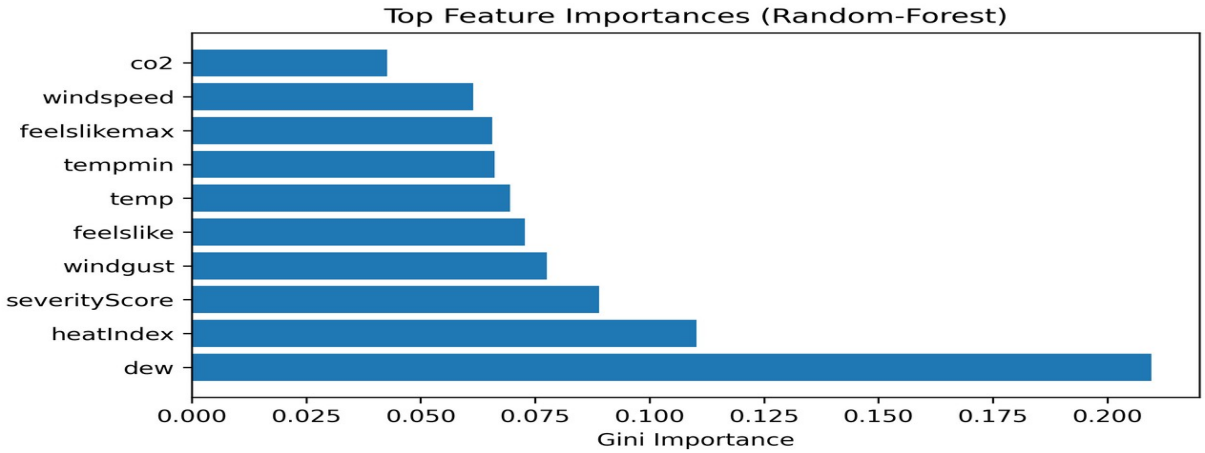


Figure 9. Top ten feature importances from the random-forest model.

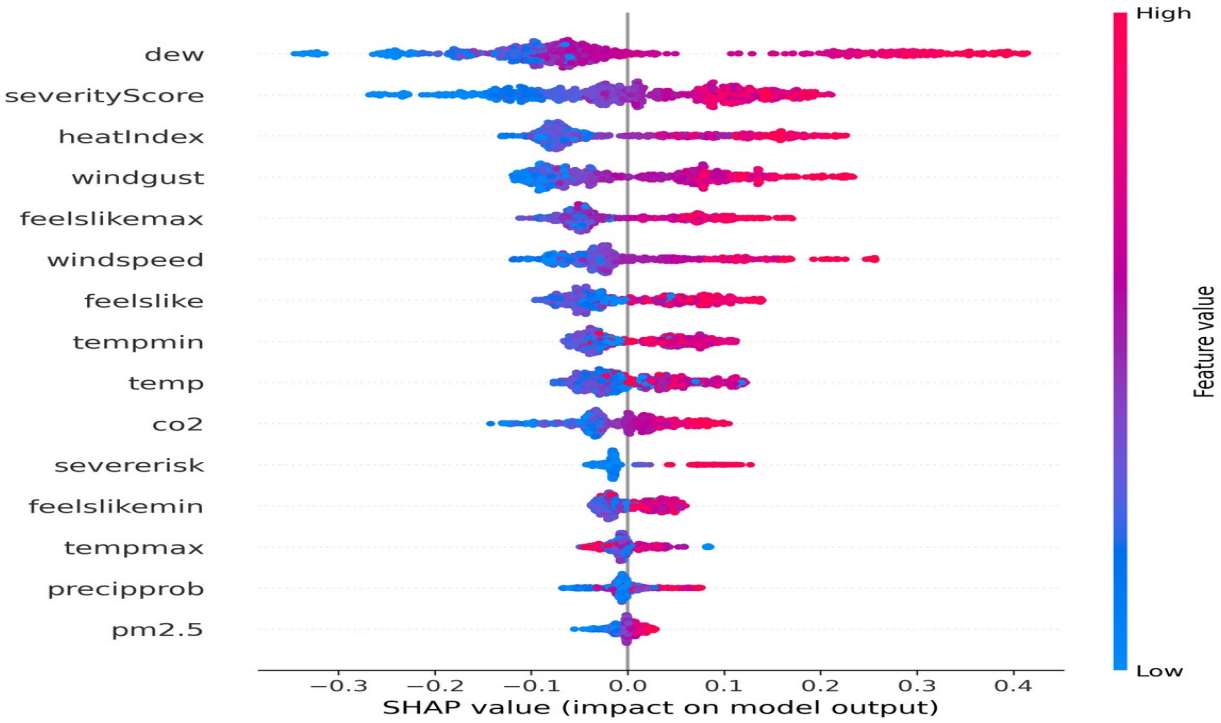


Figure 10. Shapley summary plot showing the impact of each feature on model output.

Predictive and Strategic Implications

1. **Operational triggers**.

   Use compound thresholds aligned with model behavior:

   - Advisory: Heat Index ≥ 85 and dew ≥ 65.
   - Escalation: Heat Index ≥ 90 or (Heat Index ≥ 80 and pm2.5 ≥ 40).

     These cut points map to the observed threshold region and keep PM2.5 influential when notably elevated.

2. **Targeted feature engineering & monitoring.**

   Add interactions and lags the model is signaling: dew×temp (or explicit heatIndex bins), rolling means/lags for pm2.5 (24–72h), and event flags (wildfire/industrial). Track weekly CV R^2/RMSE and SHAP drift to detect concept shift.

3. **Risk communication & planning.**

   Calibrate predictions to actionable bands (e.g., “low/medium/high”) using quantiles of the *residuals* so band edges reflect actual error. For resource planning, prioritize days forecasted in the high dew + high heat index quadrant as the model is most confident there, and the causal story is strong.